

Confidence-Bounded Physical Feasibility Ranking (CB-PFR): A Proposed Uncertainty-Aware Selection Layer for QUBO-Generated Candidates

Fahry Bagus Adiputra
gambaranfachry03@gmail.com

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Evidence-status notice. This revision incorporates the supplied executable source code, synthetic configuration/results, trial-level CSV, reproducibility script, MC/DC validation record, and QUBO–physics mapping audit. The synthetic numerical results are therefore treated as *verified from available artifacts*. The MC/DC transport runs remain execution/benchmark evidence only and are not physical validation of CB-PFR. No defensible QUBO-to-C5G7 physical mapping was found, so no reactor-physics effectiveness claim is made.

Abstract

In-core fuel-management optimization may produce multiple discrete candidate loading patterns, including candidates generated from quadratic unconstrained binary optimization (QUBO) formulations. A low QUBO objective value does not, by itself, establish physical feasibility when relevant physical quantities are evaluated after candidate generation and are subject to quantified uncertainty. This paper proposes Confidence-Bounded Physical Feasibility Ranking (CB-PFR), a post-optimization selection layer for a finite candidate pool. CB-PFR separates explicit QUBO feasibility from physical feasibility, widens estimated physical bounds using a documented uncertainty measure, aggregates normalized conservative violations, and applies a lexicographic ranking rule in which QUBO energy is a lower-priority criterion. The supplied research handover reports a synthetic experiment with 1,000 trials, 12 candidates per trial, four comparison procedures, a common observation budget, seed 20261005, and a synthetic feasibility threshold of 0.98; it records selected latent-feasibility rates of 0.857 (QUBO-only), 0.933 (point estimate), 0.976 (CB-PFR), and 0.966 (uniform margin). These values are verified against the supplied trial-level CSV and executable analysis artifacts. A separate MC/DC transport benchmark is treated as execution evidence only and is not used as physical validation of CB-PFR. The contribution is therefore methodological and conditional: the paper defines an auditable selection rule and an evaluation protocol, while leaving physical effectiveness, global novelty, and transfer beyond the specified synthetic generator as open empirical questions.

Keywords: QUBO; uncertainty quantification; candidate ranking; physical feasibility; nuclear fuel management; robust optimization; simulation-based optimization.

1 Introduction

In-core fuel management is a discrete optimization problem in which fuel assemblies are assigned to core positions subject to loading, inventory, positional, and operational constraints. Prior work has investigated QUBO and quantum-inspired formulations as candidate-generation mechanisms for fuel-reloading problems [1]. Such formulations provide a compact binary representation of a difficult combinatorial search problem. Nevertheless, the output of an optimizer is not necessarily

a complete decision rule when important physical quantities are evaluated by a subsequent simulator.

A candidate may have a favorable QUBO energy while failing a physical requirement evaluated only after simulation. Conversely, a candidate passing a point-estimate physical screen may lie close to a limit, so that uncertainty in the estimated metric changes the decision. This creates a post-optimization selection problem: given a finite pool of candidates and estimated physical metrics with associated uncertainty, which candidate should be selected while preserving an explicit priority for feasibility?

CB-PFR is proposed as a selection layer for this problem. It does not replace the QUBO optimizer, modify its Hamiltonian, or constitute a new quantum-annealing algorithm. Candidates may in principle originate from quantum annealing, simulated annealing, quantum-inspired optimization, or another candidate generator, provided they are evaluated under a common physical model.

The paper makes four bounded contributions. First, it formalizes a separation between explicit QUBO feasibility and conservative physical feasibility. Second, it defines candidate-specific operational uncertainty bounds and normalized violations. Third, it specifies a deterministic lexicographic ranking rule. Fourth, it defines comparison and verification tests that can distinguish the effect of candidate-specific uncertainty from generic conservatism. The paper does not claim reactor-safety certification, physical validation, or universal methodological superiority.

2 Related Work and Methodological Positioning

Robust optimization and chance-constrained optimization provide established ways to account for uncertainty in constrained decision problems [2]. Ranking and selection under input uncertainty likewise addresses decisions among alternatives when estimated inputs are uncertain [3]. These bodies of work establish important conceptual precedents for uncertainty-aware feasibility and selection. The present method should therefore be understood as a particular procedural combination rather than as an automatically established globally novel principle.

The connection to fuel-management optimization is supplied by QUBO formulations in which binary variables encode discrete fuel-placement decisions and penalties enforce constraints [1]. The role of CB-PFR is downstream: it receives a finite candidate pool rather than solving the full binary state space. Physical transport quantities such as k_{eff} may be supplied by a transport simulator such as OpenMC [4], but the simulator is not part of the definition of CB-PFR itself.

Table 1: Methodological positioning based on the supplied evidence.

Dimension	Related approaches	CB-PFR	Status of substantive difference
Input uncertainty	Robust/chance-constrained formulations; ranking under uncertain inputs	Candidate-specific uncertainty measure used to widen physical bounds	Combination is defined; broader novelty remains unestablished
Feasibility	Constraint satisfaction or probabilistic/robust feasibility	Explicit QUBO feasibility followed by conservative physical feasibility	Procedural distinction is explicit
Candidate-specific bounds	Depends on uncertainty model	$\hat{\mu}_{ij} \pm z_j \hat{s}_{ij}$	Requires calibration if probabilistic meaning is claimed
Violation aggregation	Common in constrained decision methods	Normalized weighted conservative violation	Useful for ranking; weights/scales are design choices
Ranking	Weighted or lexicographic multi-criteria rules	Lexicographic feasibility-first rule	Auditable ordering is the stated design rationale
Reference criterion	Problem-dependent	Independently defined F_{ref}	Necessary for empirical evaluation
Simulation budget	Problem-dependent	Same synthetic observation budget across methods in the reported design	Must be verified from raw artifacts before stronger claims

No supplied evidence establishes that every element of the procedure is individually new. The defensible claim is that CB-PFR specifies a concrete, auditable combination of these ingredients for post-optimization candidate selection.

3 Problem Formulation

Let the finite candidate pool be

$$C = \{x_1, \dots, x_{N_c}\}, \quad x_i \in \{0, 1\}^n, \quad (1)$$

where N_c is the number of candidates and n is the number of binary variables. CB-PFR ranks only this pool; it does not solve the complete binary optimization problem.

For candidate x_i , define the QUBO energy

$$H_Q(x_i) = x_i^\top Q x_i. \quad (2)$$

A penalty formulation may be written as

$$H_Q(x) = H_{\text{obj}}(x) + \sum_{r=1}^R \lambda_r P_r(x), \quad \lambda_r \geq 0. \quad (3)$$

Low energy is not assumed to be equivalent to complete feasibility. We therefore define an explicit indicator

$$I_Q(x_i) = \begin{cases} 0, & \text{if all explicitly checked QUBO constraints hold,} \\ 1, & \text{otherwise.} \end{cases} \quad (4)$$

Suppose candidate x_i has m physical metrics. Let $\hat{\mu}_{i,j}$ denote an estimated metric and let L_j and/or U_j denote documented physical limits. The limits are not defined by CB-PFR; an applied study must obtain them from a defensible specification, benchmark, or other documented source.

4 CB-PFR Method

4.1 Uncertainty and operational bounds

The uncertainty quantity $\hat{s}_{i,j}$ must be defined by the experiment. It may be a standard error, a standard deviation, a simulator-reported tally uncertainty, an input-uncertainty estimate, or another explicitly justified quantity. These quantities are not interchangeable.

Given a nonnegative multiplier z_j , define operational bounds

$$B_{i,j}^- = \hat{\mu}_{i,j} - z_j \hat{s}_{i,j}, \quad B_{i,j}^+ = \hat{\mu}_{i,j} + z_j \hat{s}_{i,j}. \quad (5)$$

These are operational conservative bounds. Without additional statistical assumptions and calibration, they are not automatically confidence intervals, prediction intervals, or statements such as “95% probability.” In particular, $z_j = 1.96$ has no universal interpretation in this framework.

For a two-sided constraint, conservative feasibility requires

$$B_{i,j}^- \geq L_j \quad \text{and} \quad B_{i,j}^+ \leq U_j. \quad (6)$$

For an upper-only constraint, require $B_{i,j}^+ \leq U_j$; for a lower-only constraint, require $B_{i,j}^- \geq L_j$.

Define

$$I_{P,j}^{\text{CB}}(x_i) = \begin{cases} 0, & \text{if the applicable conservative test passes,} \\ 1, & \text{otherwise,} \end{cases} \quad (7)$$

and

$$I_P^{\text{CB}}(x_i) = \max_{1 \leq j \leq m} I_{P,j}^{\text{CB}}(x_i). \quad (8)$$

Thus $I_P^{\text{CB}} = 0$ means that all mandatory physical tests pass under the specified operational rule; it does not certify reactor safety.

4.2 Conservative violation and normalization

For a two-sided constraint, define

$$v_{i,j} = \max(0, L_j - B_{i,j}^-) + \max(0, B_{i,j}^+ - U_j). \quad (9)$$

For one-sided constraints, retain only the relevant term. With a predeclared positive scale s_j^{scale} , define

$$\tilde{v}_{i,j} = \frac{v_{i,j}}{s_j^{\text{scale}}}, \quad (10)$$

and aggregate

$$V_{\text{CB}}(x_i) = \sum_{j=1}^m w_j \tilde{v}_{i,j}, \quad w_j \geq 0. \quad (11)$$

Normalization prevents incompatible units from dominating solely because of numerical scale. The weights and scales are decision parameters and can change ranking; therefore they should be fixed before confirmatory evaluation rather than tuned after observing outcomes.

4.3 Lexicographic selection

The proposed selection rule is

$$x_{\text{CB}}^* = \text{lex argmin}_{x_i \in C} (I_Q(x_i), I_P^{\text{CB}}(x_i), V_{\text{CB}}(x_i), H_Q(x_i), i). \quad (12)$$

The priorities are: (1) explicit QUBO feasibility, (2) conservative physical feasibility, (3) normalized conservative violation, (4) QUBO energy as a tie-breaker, and (5) candidate index as a deterministic final tie-breaker.

This ordering is a design choice intended to be auditable. A lower-priority improvement cannot compensate for a higher-priority feasibility failure. If no candidate passes all mandatory checks, the correct output is “no candidate satisfies all mandatory constraints.” A least-violating candidate may be returned for diagnosis, but must not be labeled feasible.

4.4 Algorithm

The implementation separates candidate generation from candidate selection. The QUBO module represents each loading as a binary vector and checks the declared structural constraints. The ranking module consumes a fixed candidate pool together with metric estimates and uncertainty magnitudes. This separation is important because CB-PFR is not a replacement optimizer: it is a deterministic post-optimization selection layer.

Algorithm 1 CB-PFR post-optimization candidate ranking

Require: Candidate pool $C = \{x_1, \dots, x_{N_c}\}$; QUBO model (Q, H_0) ; metrics $j = 1, \dots, m$; limits L_j, U_j ; multipliers z_j ; scales $s_j^{\text{scale}} > 0$; weights $w_j \geq 0$; estimates $(\hat{\mu}_{i,j}, \hat{s}_{i,j})$.

Ensure: Sorted list of **RankedCandidate** records.

```

1: for all  $x_i \in C$  do
2:   Validate candidate vector through the QUBO model.
3:    $I_Q(x_i) \leftarrow \mathbf{1}[x_i \text{ violates any explicit QUBO constraint}]$ .
4:    $H_Q(x_i) \leftarrow x_i^\top Q x_i + H_0$ .
5:   if ranking mode is qubo_only then
6:      $K_i \leftarrow (I_Q(x_i), H_Q(x_i), i)$  and set physical assessment to None.
7:   else
8:     if mode is point_estimate then
9:       set every metric width override to 0.
10:    else if mode is uniform_margin then
11:      use the supplied common nonnegative margin for each metric.
12:    else if mode is cb_pfr then
13:      use candidate-specific width  $z_j \hat{s}_{i,j}$ .
14:    end if
15:     $I_P^{CB}(x_i) \leftarrow 0$  and  $V_{CB}(x_i) \leftarrow 0$ .
16:    for  $j = 1$  to  $m$  do
17:      if metric estimate is missing, non-finite, or has negative uncertainty then
18:        mark metric status invalid; set violation and normalized violation to  $+\infty$ .
19:      else
20:         $b_{i,j} \leftarrow z_j \hat{s}_{i,j}$  (or the selected width override).
21:         $B_{i,j}^- \leftarrow \hat{\mu}_{i,j} - b_{i,j}$ ;  $B_{i,j}^+ \leftarrow \hat{\mu}_{i,j} + b_{i,j}$ .
22:         $v_{i,j} \leftarrow \max(0, L_j - B_{i,j}^-) + \max(0, B_{i,j}^+ - U_j)$ , retaining only the applicable term
        for one-sided limits.
23:         $\tilde{v}_{i,j} \leftarrow v_{i,j} / s_j^{\text{scale}}$ .
24:         $V_{CB}(x_i) \leftarrow V_{CB}(x_i) + w_j \tilde{v}_{i,j}$ .
25:        if  $v_{i,j} > 0$  then  $I_P^{CB}(x_i) \leftarrow 1$ .
26:      end if
27:    end if
28:  end for
29:   $K_i \leftarrow (I_Q(x_i), I_P^{CB}(x_i), V_{CB}(x_i), H_Q(x_i), i)$ .
30: end if
31: Store candidate, assessments, energy, input index, and  $K_i$  as a RankedCandidate.
32: end for
33: Sort all stored records by ascending lexicographic  $K_i$ .
34: Return the sorted records; the selected candidate is the first record.

```

The pseudocode above is deliberately aligned with the supplied `ranking.py`. In that implementation, missing or invalid physical estimates produce infinite violation and a failed physical indicator; they are not silently discarded. The four ranking modes are implemented explicitly. For CB-PFR, the physical part of the key is (I_P, V) , followed by QUBO energy and the original input index. Python’s stable sort is applied to these explicit tuple keys, so the final candidate order is deterministic for a fixed candidate sequence. The helper `selected_is_conservatively_feasible` separately checks whether the selected record has both $I_Q = 0$ and a physical indicator of zero. Therefore, the algorithm itself returns the ranked list; it does not silently relabel a least-violating candidate as physically feasible when no candidate passes.

Important distinction from the manuscript-level decision rule. Equation (12) describes the intended CB-PFR selection policy. The executable implementation is more precise about edge cases: the physical indicator and aggregate violation are calculated even when an estimate is missing or invalid, using $+\infty$ violation; the first sorted candidate is still returned as a diagnostic selection. A release-level caller must use `selected_is_conservatively_feasible` before reporting that selection as passing. This distinction prevents the paper from attributing a stronger fail-closed behavior to code that only implements the ranking primitive.

4.4.1 Candidate-generation algorithm

Before Algorithm 1 can operate, the QUBO module constructs a candidate representation with three binary variables per position (fresh, once-burned, twice-burned in the toy formulation), expands the declared penalty terms into an upper-triangular QUBO matrix plus a constant offset, and can enumerate/check feasible binary vectors. The supplied implementation explicitly evaluates cell-loading, inventory, forbidden-region, adjacency, and optional symmetry constraints. This stage produces the candidate pool; CB-PFR does not alter these constraints.

Algorithm 2 QUBO candidate construction and screening in the supplied toy model

Require: Core specification with positions, regions, edges, inventories, and optional symmetry pairs.

Ensure: Binary candidate set C and QUBO energy function H_Q .

- 1: Create three binary variables $z_{b,p}$ for each burnup category b and position p .
 - 2: Add exactly-one penalty for every position.
 - 3: Add inventory-count penalties for all three categories.
 - 4: Add forbidden-region penalties for prohibited category–region combinations.
 - 5: Add declared adjacency penalties for prohibited neighboring categories.
 - 6: Add optional symmetry penalties for declared position pairs.
 - 7: Expand all square penalties into the upper-triangular matrix Q and constant offset H_0 .
 - 8: Enumerate or receive candidate binary vectors.
 - 9: **for all** candidate vectors x_i **do**
 - 10: Check binary dimension and all explicit constraints.
 - 11: Compute $H_Q(x_i) = x_i^\top Q x_i + H_0$.
 - 12: **end for**
 - 13: Return the QUBO model and the feasible candidate pool.
-

4.4.2 Computational complexity

For a fixed candidate pool, Algorithm 1 requires $O(N_c m)$ metric operations plus QUBO-feasibility/energy checks and $O(N_c \log N_c)$ sorting. Thus, when simulator outputs are already available, the ranking stage is approximately

$$O(N_c(C_Q(n) + m)) + O(N_c \log N_c), \quad (13)$$

where $C_Q(n)$ denotes the cost of one explicit QUBO-constraint/energy check. In an end-to-end workflow, physical simulation and uncertainty estimation may dominate the computational cost.

5 Experimental Design

The supplied release package contains the executable ranking/QUBO source, the reduced synthetic calibration script, the frozen synthetic configuration, the aggregate result CSV, and the 4,000-row trial-level CSV. These artifacts are sufficient to audit and recompute the reported aggregate rates and paired trial-level statistics from the recorded outputs. They should therefore

be classified as *verified from supplied artifacts*. This status does not mean that the full 1,000-trial generation was independently rerun in the present audit environment: the packaged reproduction script expects the repository layout and auxiliary modules described by its own paths, and that complete execution environment is not claimed here. The result is therefore not labeled *independently reproduced*.

The comparison procedures are:

- A. QUBO-only:** rank using explicit QUBO feasibility and QUBO energy without physical screening.
- B. Point estimate:** apply physical limits directly to $\hat{\mu}$ without uncertainty widening.
- C. CB-PFR:** use candidate-specific operational bounds, conservative feasibility, normalized violations, and the lexicographic rule.
- D. Uniform margin:** use a common predeclared margin rather than candidate-specific uncertainty.

An independent reference criterion should be defined before evaluation. Let $F_{\text{ref}}(x_i) \in \{0, 1\}$ denote reference feasibility. For trial t , let $x_{M,t}^*$ be the candidate selected by method M ; the selected-candidate reference-feasibility rate is

$$R_M = \frac{1}{T} \sum_{t=1}^T F_{\text{ref}}(x_{M,t}^*). \quad (14)$$

This rate is an empirical statistic over the specified test cases, not a probability of real-world reactor safety.

A defensible evaluation should additionally report false rejection among reference-feasible candidates, rejection rates, calibration if probabilistic interpretation is claimed, sensitivity to z_j , stability across repeated candidate-generation runs, and separate generation, simulation, uncertainty-estimation, ranking, and total runtime.

6 Synthetic Experimental Results

6.1 Reported aggregate results

The handover records the following selected latent-feasibility rates:

Table 2: Synthetic results recorded in the supplied research handover.

Method	Selected latent-feasibility rate
QUBO-only	0.857
Point estimate	0.933
CB-PFR	0.976
Uniform margin	0.966

These numbers support a narrowly worded statement: *under the specified synthetic design as reported by the handover, CB-PFR had the highest recorded selected latent-feasibility rate among the four procedures*. They do not establish physical effectiveness, reactor safety, or universal superiority. The supplied `per_seed_results.csv`, `main_results.csv`, synthetic runner, configuration, and ranking implementation provide an auditable computational evidence chain for these rates.

The absolute differences in the recorded aggregate rates are 0.043 versus point estimate, 0.119 versus QUBO-only, and 0.010 versus uniform margin. The small gap against uniform margin is particularly relevant: it motivates testing whether candidate-specific uncertainty contributes information beyond generic conservatism.

6.2 Paired comparisons reported by the handover

The supplied trial-level records give the following exclusive successes: CB-PFR versus point estimate, 47 versus 4; CB-PFR versus QUBO-only, 125 versus 6; and CB-PFR versus uniform margin, 12 versus 2. Using these discordant-pair counts in an exact two-sided sign test gives p-values of approximately 2.42×10^{-10} , 4.82×10^{-30} , and 1.29×10^{-2} , respectively. The paired counts and corresponding trial-level records are available in the supplied `per_seed_results.csv`; these values are recomputed from the available evidence rather than taken only from a summary table.

For the sign test, the null hypothesis is that, conditional on a discordant pair, either method is equally likely to be the better method. Ties are excluded from the discordant-pair count. The analysis unit is the trial pair, not an individual candidate. Because three pairwise comparisons are considered, multiplicity should be addressed in a confirmatory analysis. A Bonferroni-adjusted threshold of $0.05/3 \approx 0.0167$ would still classify the three reported p-values as below threshold, but this statement remains conditional on the correctness of the reported discordant counts and the validity of the sign-test setup.

Statistical significance in this synthetic setting does not establish algorithmic superiority outside the generator. Effect sizes, confidence intervals, and sensitivity to generator assumptions remain necessary for a stronger empirical characterization.

6.3 Pilot results and negative evidence

The handover states that a 24-trial pilot exists and should not be hidden merely because it may be less favorable. The pilot data themselves were not supplied in the present evidence package, so this revision does not reproduce or numerically reinterpret them. A final computational release should report the pilot protocol separately if its design differs from the 1,000-trial experiment. Results from different protocols should not be pooled without justification.

7 MC/DC Benchmark Study and Its Scope

The release’s physical-simulation audit records seven MC/DC runs on the C5G7-TD benchmark without control rods. One recorded run has

$$k_{\text{mean}} = 1.1656485666488008, \quad (15)$$

$$k_{\text{sdev}} = 0.016899436969674532, \quad (16)$$

with 12,000 histories. These values are reported as recorded execution evidence; the accompanying physical-simulation audit documents direct inspection of the available HDF5 outputs and classifies the evidence as software verification plus limited benchmark comparison, not CB-PFR physical validation.

The distinction between k_{sdev} and a standard error is essential. The supplied evidence does not establish the simulator convention sufficiently for this paper to relabel the value as a standard error, confidence interval, or calibrated uncertainty. The reported reference value of approximately 1.16562 is likewise retained only as a reported comparison because its external source was not independently verified in the supplied artifacts.

Most importantly, this benchmark is not a physical validation of CB-PFR. The supplied handover explicitly states that OpenMC was not run for the CB-PFR candidate evaluation, no

complete OpenMC model was found for that purpose, and no defensible mapping was established from categories such as fresh, once-burned, and twice-burned to material composition, burnup state, geometry, density, temperature, and nuclear data. Consequently, no claim is made that QUBO candidates selected by CB-PFR have been physically validated.

8 Discussion

8.1 Why candidate-specific uncertainty can help

Point-estimate screening treats candidates with the same estimated metric similarly when their uncertainty differs. Candidate-specific conservative bounds can distinguish a candidate whose estimate is comfortably inside a limit from another whose estimate is near the same limit but substantially more uncertain. This can improve selection when the uncertainty measure is informative and calibrated enough for the operational decision rule.

8.2 When uniform margins can match or outperform CB-PFR

A uniform margin can perform similarly when candidate uncertainties are approximately homogeneous, when the candidate-specific uncertainty estimates contain little useful information, or when the uniform margin happens to approximate the relevant conservative protection. If candidate-specific uncertainty is noisy or systematically miscalibrated, using it can even increase false rejection or otherwise degrade selection. These are testable empirical conditions rather than assumptions that favor CB-PFR by construction.

8.3 Effect of normalization and weights

Normalization and weights matter whenever more than one physical constraint can be violated. Changing s_j^{scale} changes the numerical contribution of a metric; changing w_j changes its priority inside V_{CB} . Thus these parameters are not innocuous. Their values should be specified independently of the final performance comparison, and sensitivity analysis should be reported.

8.4 Interpretation of the lexicographic rule

The lexicographic key is mathematically well defined for a finite candidate set provided all components are finite and candidate indices are unique. Its semantics are transparent: a lower-priority objective cannot compensate for a higher-priority failure. This is preferable to an arbitrary weighted sum when the scientific decision policy is explicitly feasibility-first. However, the ordering is a design decision and should be justified by the application rather than presented as a universal optimality theorem.

8.5 Computational versus physical evidence

The synthetic experiment tests the selection mechanism under a controlled generator. The MC/DC benchmark tests transport-code execution on a benchmark configuration. These are complementary but different forms of evidence. Neither substitutes for an experiment that connects actual QUBO fuel-loading candidates to a documented physical model and evaluates whether the selection rule improves the intended physical decision criterion.

9 Limitations and Threats to Validity

The principal limitation is evidentiary and reproducibility-related. The current release contains the paper source, executable ranking/QUBO/synthetic-boundary modules, configuration and

recorded CSV outputs, and artifact-audit script. These files support direct audit of the recorded synthetic results, but the complete archived experiment runner is not turnkey in this release: it expects additional repository components (including an experiment-utility module and a toy-core configuration) that are not present here. Accordingly, the recorded numerical results are verified from supplied artifacts, while a fresh end-to-end 1,000-trial regeneration is not claimed.

The synthetic generator is another central limitation. Even a correctly executed result on a known-normal synthetic model is conditional on that model. Distributional mismatch, uncertainty miscalibration, heterogeneous constraints, and different candidate-generation mechanisms may change the ranking behavior.

No physical QUBO-to-transport mapping is established. In particular, no defensible physical interpretation of the fuel-category variables sufficient for candidate validation is supplied. Therefore, the paper makes no claim of reactor-safety validation or physical effectiveness.

The novelty claim is also deliberately limited. Robust optimization, chance constraints, ranking and selection under uncertainty, and lexicographic decision procedures are established methodological areas. The specific combination proposed here may be useful, but global methodological novelty has not been established by the supplied evidence.

10 Evidence and Verification Status

Table 3 distinguishes the three evidence levels that are most easily conflated in a computational research release.

Table 3: Evidence-status taxonomy used in this release.

Status	Meaning in this paper
<i>Verified from supplied artifacts</i>	A numerical or methodological statement is traceable to the released source/-configuration/output artifacts and can be recomputed or inspected from those artifacts. This status does not imply a fresh end-to-end rerun.
<i>Independently reproduced</i>	The released computational environment has been executed from the specified inputs to regenerate the reported result independently of the archived output. This stronger status is not claimed for the 1,000-trial study in the present audit.
<i>Recorded execution evidence</i>	A run was recorded and documented, but its scope is limited to the executed benchmark/software configuration. The MC/DC C5G7 study belongs here.
<i>Physically validated</i>	The method’s actual research hypothesis has been tested using a defensible physical model and candidate-to-physics mapping. CB-PFR is not at this status.

11 Reproducibility and Data Availability

The current release package contains the paper source and bibliography, the QUBO/ranking/synthetic/OpenMC-boundary source files, the reduced synthetic configuration and outputs, the trial-level CSV, the physical-simulation audit, the QUBO-physics mapping audit, and a reproduction script. The aggregate and trial-level CSVs are therefore available for artifact-level audit, and the reported rates and paired counts can be recomputed directly from the supplied CSV.

The stronger label *independently reproduced* is not used. The archived reproduction script expects a repository layout containing a `src/cbpfr` package, tests, auxiliary experiment utilities, configuration files, and physical-analysis scripts beyond the flat artifact subset included with this paper package. Because that complete execution environment was not independently executed in the present audit, the release makes no claim that the 1,000-trial experiment has been rerun from scratch here.

This distinction is intentional: *verified from supplied artifacts* means that the claim is traceable to files available in the release; *independently reproduced* would require a fresh execution from the released computational environment; and *physically validated* would additionally require a defensible mapping from QUBO candidates to a physical transport/depletion model. The third condition is explicitly not satisfied.

12 AI-Assistance Disclosure

AI-assisted tools were used to help organize, edit, and formalize parts of the manuscript and the proposed procedure. The author remains responsible for checking the mathematical formulation, references, implementation, results, and final scientific claims.

13 Conclusion

This paper defines Confidence-Bounded Physical Feasibility Ranking as a post-optimization selection layer for a finite pool of QUBO-generated candidates. The method separates explicit QUBO feasibility from conservative physical feasibility, uses uncertainty-widened operational bounds, aggregates normalized violations, and applies a feasibility-first lexicographic ranking rule.

The supplied release artifacts support an auditable synthetic result: CB-PFR has the highest selected latent-feasibility rate among the four compared procedures (0.976 versus 0.933, 0.857, and 0.966) under the specified synthetic generator and recorded evaluation protocol. The underlying CSV and source/configuration artifacts are available in the release, so this result is classified as *verified from supplied artifacts*. It is not labeled independently reproduced because the complete archived execution environment was not rerun in the present audit.

The separate MC/DC benchmark remains recorded transport-execution evidence and does not validate CB-PFR physically. The QUBO–physics audit reports no defensible mapping from the abstract `fresh/once_burned/twice_burned` categories to a physical transport model. Accordingly, the appropriate current claim is methodological and conditional: CB-PFR is a precisely specified, auditable post-optimization ranking layer with positive evidence on the supplied synthetic benchmark, while physical effectiveness, transfer beyond the synthetic generator, and global methodological novelty remain open questions.

A stronger future study should release the complete computational artifacts, independently reproduce the synthetic results, verify statistical analyses from trial-level data, calibrate the uncertainty measures, establish a defensible QUBO-to-physics mapping, and evaluate selected candidates under an independently specified physical reference criterion.

Notation Summary

Symbol	Meaning
C, N_c	Candidate pool and number of candidates
x_i, n	Candidate binary vector and number of binary variables
$Q, H_Q(x_i)$	QUBO matrix and QUBO energy
$I_Q(x_i)$	Explicit QUBO-feasibility indicator; 0 means feasible
m, j	Number of physical metrics and metric index
$\hat{\mu}_{i,j}$	Estimated physical metric value
$\hat{s}_{i,j}$	Uncertainty measure used by CB-PFR
z_j	Uncertainty multiplier
$B_{i,j}^-, B_{i,j}^+$	Conservative lower and upper operational bounds

L_j, U_j	Lower and upper physical limits
$I_P^{\text{CB}}(x_i)$	Conservative physical-feasibility indicator
$v_{i,j}, \tilde{v}_{i,j}$	Conservative and normalized violation
w_j, s_j^{scale}	Aggregation weight and normalization scale
$V_{\text{CB}}(x_i)$	Aggregate conservative violation
$F_{\text{ref}}(x_i)$	Independently defined reference-feasibility indicator
R_M	Selected-candidate reference-feasibility rate for method M

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